

Data Science Program, George Washington University

Sample Midterm Exam

DATS 6101 (Introduction to Data Science)

Date: **October 26, 2026**

Instructor: Prof. Sushovan Majhi

Name: _____

Please read the following instructions carefully.

- This is a **sample paper** for practice only, at the real length, the real shape and the real difficulty, covering everything through Oct 19.
- It prints **30 problems**. **Answer any 25**. Each is worth **6 points**, so the paper is marked out of **150**, and if you answer more than 25 **your best 25 count**.
- You have **2 hours 30 minutes**, the full session.
- Most problems are multiple choice and **exactly one option is correct**. Mark it clearly (✓). The rest say what they want.
- The exam is **open-book** and **open-note**, but **printed material only**: the chapters covered, the labs whose solutions have been released, the homework covered, and your own notes.
- **Laptops, tablets, phones** and **watches** must be **switched off and put away**. No AI assistance and no communication of any kind.
- Nothing has to be computed that cannot be done on paper: every critical value a question needs is **printed in the question**. A correct number with no sentence around it earns part marks.
- All notation has its usual meaning. You may use standard results directly by stating them appropriately. **Read the instructions for individual questions carefully** before answering.

Question	Points	Score
1	6	
2	6	
3	6	
4	6	
5	6	
6	6	
7	6	
8	6	
9	6	
10	6	
11	6	
12	6	
13	6	
14	6	
15	6	
16	6	
17	6	
18	6	
19	6	
20	6	
21	6	
22	6	
23	6	
24	6	
25	6	
26	6	
27	6	
28	6	
29	6	
30	6	
Total:	180	

Part A Reading a claim

1. (6 points) A study reports mean satisfaction **7.2 (95% CI 6.9 to 7.5)**. Which statement is defensible?
 - ☐ 95% of respondents rated between 6.9 and 7.5
 - ☐ There is a 95% probability the true mean is between 6.9 and 7.5
 - ☐ If the study were repeated many times, about 95% of intervals built this way would contain the true mean
 - ☐ The next respondent will rate between 6.9 and 7.5 with 95% probability
2. (6 points) Exam scores have mean 72 and standard deviation 11, from $n = 400$ students, so the standard error is 0.55. A report says “*almost every student scored within about half a point of 72.*” The error is
 - ☐ describing students with a number that describes the average of 400 of them
 - ☐ using the standard error instead of the variance
 - ☐ forgetting to multiply by 1.96
 - ☐ using $n = 400$ when the class is larger
3. (6 points) A comparison returns $p = 0.30$. The correct reading is
 - ☐ there is a 30% chance the null hypothesis is true
 - ☐ the two groups are equal
 - ☐ data at least this extreme are unremarkable if the null is true, which leaves both “no effect” and “too small a study” alive
 - ☐ the study has 30% power
4. (6 points) An app’s rating is estimated from **40,000** reviews, all left voluntarily. The interval is extremely narrow. This tells you
 - ☐ the rating is accurate to within the interval
 - ☐ the rating is precise, and says nothing about whether reviewers resemble users
 - ☐ the sample is large enough for the CLT, so the estimate is unbiased
 - ☐ nothing, because ratings are ordinal
5. (6 points) X and Y are independent. Then $\text{Var}(X - Y)$ equals
 - ☐ $\text{Var}(X) - \text{Var}(Y)$
 - ☐ $\text{Var}(X) + \text{Var}(Y)$
 - ☐ $\text{Var}(X) \text{Var}(Y)$
 - ☐ it cannot be determined without the covariance

Part B Probability and random variables

6. (6 points) A condition affects **2%** of a population. A test detects it in 95% of those who have it and is negative in 90% of those who do not. Someone tests positive. The probability they have the condition is closest to
- ☐ 95%
 - ☐ 50%
 - ☐ 16%
 - ☐ 2%
7. (6 points) X has variance 9. Then $\text{Var}(3X - 2)$ is
- ☐ 25
 - ☐ 27
 - ☐ 81
 - ☐ 79
8. (6 points) Which is a valid probability density on $[0, 3]$?
- ☐ $f(x) = 2x/9$
 - ☐ $f(x) = x/3$
 - ☐ $f(x) = 1/2$
 - ☐ $f(x) = 3 - x$
9. (6 points) Of 500 site visits, 200 came from search and 300 from social; 40 of the search visits and 60 of the social visits converted. Converting and arriving from search are
- ☐ independent — the conversion rate is 20% in both groups
 - ☐ dependent, because social produced more conversions
 - ☐ dependent, because the groups are different sizes
 - ☐ impossible to judge without a significance test
10. (6 points) Two fair dice are rolled. The events “*the first shows a six*” and “*the total is seven*” are
- ☐ independent, because the total is equally likely to be 7 whatever the first die shows
 - ☐ dependent, because a six makes a large total more likely
 - ☐ mutually exclusive
 - ☐ dependent, because both involve the first die

11. (6 points) **Short answer.** A delivery fee is \$0 with probability 0.5, \$5 with probability 0.3 and \$20 with probability 0.2. Compute $E[X]$ and $\text{Var}(X)$, showing the working.

Part C Expectation, variance and the CLT

12. (6 points) A study multiplies its sample size by **nine**. The standard error of the mean is multiplied by
- ☐ 1/9
 - ☐ 1/3
 - ☐ 3
 - ☐ it is unchanged
13. (6 points) A clustered survey of **800** has a true standard error **twice** what $s/\sqrt{800}$ reports. Its effective sample size is about
- ☐ 400
 - ☐ 200
 - ☐ 800
 - ☐ 1,600
14. (6 points) The central limit theorem says nothing about which of these?
- ☐ the sampling distribution of $\bar{X}$ at large n
 - ☐ the sampling distribution of the sample *median*
 - ☐ the mean of the sampling distribution of $\bar{X}$
 - ☐ the spread of the sampling distribution of $\bar{X}$
15. (6 points) The population is exponential, whose skewness is 2. You average samples of $n = 25$. The skewness of the sample means is closest to
- ☐ 2
 - ☐ 1
 - ☐ 0.4
 - ☐ 0

16. (6 points) For a Cauchy population, the mean of 1,000 draws is no more precise than the mean of 10. This is because
- ☐ 1,000 is still too small
 - ☐ the Cauchy has no finite variance, so $\sigma/\sqrt{n}$ has nothing to describe
 - ☐ the draws are dependent
 - ☐ the Cauchy is discrete
17. (6 points) **Short answer.** A population has $\sigma = 24$. You draw $n = 36$. Give the standard error of $\bar{X}$, and one sentence saying what that number describes — being explicit about how it differs from σ .

Part D Confidence intervals

18. (6 points) A 95% interval for a mean from $n = 25$ uses which critical value?

<code>qt(0.95, 24)</code>	1.711	<code>qt(0.975, 24)</code>	2.064
<code>qt(0.975, 25)</code>	2.060	<code>qnorm(0.975)</code>	1.960

- ☐ `qt(0.95, 24)`
 - ☐ `qt(0.975, 24)`
 - ☐ `qt(0.975, 25)`
 - ☐ `qnorm(0.975)`
19. (6 points) **Short answer.** A sample of 100 tickets has $\bar{x} = 52$ and $s = 20$ minutes. Using $t_{0.975,99} = 1.984$, give the standard error and the 95% interval, with the working.
20. (6 points) A paper reports a mean of 12.4 with a 95% CI of 11.6 to 13.2 and a standard deviation of 4.0, but never gives the sample size. It is closest to

- ☐ 25
 - ☐ 96
 - ☐ 250
 - ☐ 400
21. (6 points) Reporting a **90%** interval instead of a 95% one, everything else fixed, multiplies the width by about
- ☐ 1.19
 - ☐ 0.84
 - ☐ 0.90
 - ☐ 0.95
22. (6 points) A trial converts **7 of 250**. `prop.test(7, 250, correct = FALSE)` returns an interval of 0.0136 to 0.0567 around $\hat{p} = 0.028$. It is **not** symmetric about $\hat{p}$ because
- ☐ `prop.test` applies a continuity correction
 - ☐ a rate near zero has more room to be larger than smaller, and the Wilson interval reflects that
 - ☐ the sample size is too small for symmetry
 - ☐ of rounding in the endpoints
23. (6 points) Group A has mean 20.0 and group B 24.0, each with standard error 1.5. Using 1.96 and $SE_{\text{diff}} = \sqrt{1.5^2 + 1.5^2} = 2.12$, the 95% interval for the difference is
- ☐ -0.16 to 8.16, which includes zero
 - ☐ 1.06 to 6.94, which excludes zero
 - ☐ 0.94 to 7.06, which excludes zero
 - ☐ not computable, because the intervals overlap
24. (6 points) **Short answer.** Your 95% interval for a mean difference is [0.4, 9.6]. Write the one sentence you would put in a report, and say in a second sentence what a reader may **not** conclude from it.

Part E Testing

25. (6 points) Two independent groups of 60 rate a feature from 1 to 10 and you want to know whether the average rating differs. The default choice is
- ☐ a two-sample (Welch) t-test
 - ☐ a paired t-test
 - ☐ a chi-squared test
 - ☐ a one-sample t-test on the difference
26. (6 points) You have one sample of 800 users answering yes or no, and want an interval for the share saying yes. Use
- ☐ `t.test`
 - ☐ `prop.test(x, 800, correct = FALSE)`
 - ☐ `chisq.test`
 - ☐ the Wald formula, since n is large
27. (6 points) An experiment on 2,000 sessions per arm returns $p = 0.002$ and a 95% confidence interval for the lift of **0.1% to 0.4%**. The right summary is
- ☐ a large, clearly valuable improvement
 - ☐ a difference that is hard to attribute to chance and too small to be worth much — significance and importance are different questions
 - ☐ an inconclusive result, since the interval is narrow
 - ☐ a 99.8% probability the new arm is better
28. (6 points) **Short answer.** A colleague reports “*no significant difference* ($p = 0.6$, $n = 18$ *per group*)” and proposes dropping the feature. In two sentences, say what is wrong with the inference and what you would ask for.

Part F Experiments

29. (6 points) Randomly assigning users to two versions buys you
- ☐ groups alike in expectation on everything, measured or not, so a difference in outcome can be attributed to the version

- ☐ identical groups
- ☐ a larger effective sample size
- ☐ freedom from the need for a confidence interval

30. (6 points) **Short answer.** In a randomized trial, 30% of the treatment arm stops using the product before the outcome is measured, against 10% of the control arm. In one or two sentences, say why the randomization no longer protects the comparison.

*Thirty problems printed; your best twenty-five count.
A number with no working earns part marks; the sentence is what is being graded.*